INCH-POUND
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PERFORMANCE SPECIFICATION

HEAT EXCHANGER, NBC SYSTEM GENERAL SPECIFICATION FOR

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the design, performance, inspection, and acceptance requirements for the heat exchangers used in Nuclear Biological Chemical (NBC) system of the M1A1 and M1A2 tanks and its three major components: heat exchanger, condenser and precooler.

1.2 Classification. Heat exchanger for the NBC system of the M1A1 and M1A2 tanks consists of the following (see 3.6):

Component	Drawing
Heat exchanger	12336771 (for interface only)
Condenser	12336772 (for interface only)
Precooler	12336768 (for interface only)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 4130

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-PRF-62411(AT)/1A - Heat Exchanger, NBC System, Heat Exchanger.
MIL-PRF-62411(AT)/2A - Heat Exchanger, NBC System Condenser.
MIL-PRF-62411(AT)/3A - Heat Exchanger, NBC System Precooler.

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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DRAWINGS

ARMY

12336768	- Precooler (for interface only).
12336771	- Heat Exchanger (for interface only).
12336772	- Condenser (for interface only).

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQC Z1.4	- Sampling Procedures and Tables for Inspection by Attributes (DoD Adopted).
ANSI/NCSL Z540-1	- General Requirements for Calibration Laboratories and Measuring and Test Equipment.
ANSI/SAE AS478	- Identification Marking Methods (DoD Adopted).

(Application for copies may be obtained from the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.)

NUCLEAR REGULATORY COMMISSION (NRC)

Code of Federal Regulations (CFR)	- Title 10, Part 30.

(Copies of the Code of Federal Regulations (CFR) are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets MIL-PRF-62411(AT)/1A, MIL-PRF-62411(AT)/2A and MIL-PRF-62411(AT)/3A. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.4.

3.3 Materials. Materials shall be uniform and free from imperfections or defects which affect their performance and serviceability. All non-metallic materials shall be inherently fungistatic. Asbestos, magnesium, and radioactive material shall not be used in this item. Radioactive material is defined by CFR , Title 10, Part 30, and other radioactive material in which the specific activity is greater than 0.002 microcuries per gram or the activity per item equals or exceeds 0.01 microcuries (see 4.6.1).

3.3.1 Dissimilar metals. The use of dissimilar metals in intimate contact with each other shall be avoided unless suitably protected against electrolytic corrosion (see 4.6.1).

3.3.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Design and construction. The heat exchanger shall be constructed to the form and dimensions depicted on the drawings specified on the applicable specification sheet (see 4.6.2).

3.4.1 Weight. See applicable specification sheet.

3.4.2 Interchangeability. The heat exchanger shall be designed and fabricated so that it shall be dimensionally and functionally interchangeable (see 4.6.1).

3.5 Performance.

3.5.1 Leakage.

3.5.1.1 External leakage. With the hot air passages (see 4.6), pneumatically pressurized as specified in applicable specification sheet, and with the heat exchanger immersed in water there shall be no escaping bubbles that would indicate air leakage from the weld seams, pans, or side plates. Bubbles emerging from the cold air passages (see 6.4) are acceptable (see 4.6.4.1).

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3.5.1.2 Internal leakage. With the hot air passages pressurized as specified at 73 ±18 degrees Fahrenheit (°F) (23 ±8 degrees Celsius (°C)) and isolated from the pressure source, the pressure loss in the first 20 seconds following isolation shall conform to the applicable specification sheet (see 4.6.4.2).

3.5.2 Proof pressure. The heat exchanger hot air passages shall withstand without damage or permanent deformation an air pressure as specified in the applicable specification sheet. The unit shall meet the requirements of 3.5.1 after subjection to the specified proof pressure (see 4.6.5).

3.5.3 Burst pressure. The heat exchanger hot air passages shall withstand without rupture an air pressure and temperature as specified in the applicable specification sheet (see 4.6.6).

3.5.4 Heat transfer. Heat transfer performance in terms of hot air effectiveness shall be not less than that shown in figure 1, in the applicable specification sheet (see 4.6.7).

3.5.5 Air pressure loss. Isothermal air pressure loss across the hot air passages shall not exceed the limit shown in figure 2, in the applicable specification sheet. The pressure drop presented in figure 2 is total-to-total, corrected to standard air density (see 4.6.8).

3.5.6 Environmental conditions.

3.5.6.1 Temperature.

3.5.6.1.1 High temperature. The heat exchanger shall meet the requirements of 3.5.1 after exposure to an operating temperature of 125°F (52°C). The unit shall also meet the requirements of 3.5.1 after exposure to a nonoperating storage temperature of 160°F (71°C) (see 4.6.9.1).

3.5.6.1.2 Low temperature. The heat exchanger shall meet the requirements of 3.5.1 after exposure to an operating temperature of minus (-) 60°F (-51°C). The unit shall also meet the requirements of 3.5.1 after exposure to a nonoperating storage temperature of -60°F (see 4.6.9.1).

3.5.6.2 Humidity. The heat exchanger unit shall meet the requirements of 3.5.1 after exposure to a relative humidity up to 100 percent (see 4.6.9.2).

3.5.6.3 Sand. The heat exchanger shall meet the requirements of 3.5.1 after exposure to sand particles of 0.01 to 1 millimeter (mm) diameter (see 4.6.9.3).

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3.5.6.4 Dust. The heat exchanger shall meet the requirements of 3.5.1 after exposure to dust laden air with dust particles (see 4.6.9.4).

3.5.6.5 Salt fog. The heat exchanger shall meet the requirements of 3.5.1 after exposure to a salt fog solution for a period of 48 hours (see 4.6.9.5).

3.5.6.6 Chemicals. The heat exchanger shall meet the requirements of 3.5.1 after exposure to vapors of direct contact with chemicals (see 4.6.9.6).

3.5.6.7 Cleaning spray. The heat exchanger shall meet the requirements of 3.5.1 after exposure to a cleaning spray (see 4.6.9.7).

3.5.6.8 Fungus. The heat exchanger shall not support fungus growth when exposed to specimen inoculation by spraying external surfaces with spore suspension (see 4.6.9.8).

3.5.6.9 Vibration. The heat exchanger unit shall meet the requirements of 3.5.1 after exposure to sinusoidal vibration (see 4.6.9.9).

3.5.6.10 Basic shock. The heat exchanger shall meet the requirements of 3.5.1 after exposure to three shock impulses (see 4.6.9.10).

3.5.6.11 Gun firing shock. The heat exchanger shall meet the requirements of 3.5.1 after exposure to shock impulses (see 4.6.9.11).

3.5.6.12 Ballistic shock. The heat exchanger shall meet the requirements of 3.5.1 after subjection to ballistic shock impulses (see 4.6.9.12).

3.6 Cross-reference to engineering and logistical support part number. Refer to table I for engineering and logistical support part number cross-reference interchangeability.

TABLE I. Cross-reference for part number interchangeability.

Army part and number 1/	General Dynamics part number	NSN	PIN
Heat exchanger 12336771	SC12336771	4130-01-217-8295	M62411/1
Condenser 12336772	SC12336772	2540-01-217-3217	M62411/2
Precooler 12336768	SC12336768	2540-01-185-0720	M62411/3

1/ For interface only

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3.7 Marking. The heat exchanger shall be marked in accordance with ANSI/SAE AS478.

3.8 Workmanship. Workmanship shall be such as to provide a heat exchanger, including accessories, equipment, and loose parts, free from defects, such as: cracked, dented, loose, broken, or corroded parts/components (see 4.6.2).

4. VERIFICATION

4.1 Inspection equipment. Inspection equipment must be capable of repetitive measurements to an accuracy of 10 percent of the measurement tolerance. Calibration of inspection equipment shall be in accordance with ANSI/NCSL Z540-1.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Conformance inspections (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature: 73 ±18°F (23 ±8°C).
b. Barometric pressure: 28.5 (+2, - 3) inches (in.) of mercury (Hg) (96.5 (+6.8, -10.2) kilopascals (kPa)).
c. Relative humidity: 50 ±30 percent.

4.4 First article inspection. First article inspections shall be performed on preproduction or initial production samples as specified herein. Approval of the first article sample by the Government shall not relieve the contractor of the obligation to supply heat exchangers that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.4.1 Preproduction inspection. When specified (see 6.2), the preproduction sample shall consist of 3 units and shall be inspected in the sequence listed in table II and as specified in table III.

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4.4.2 Initial production inspection. Unless otherwise specified (see 6.2), the Government shall select three from the first 200 heat exchanger units produced under the production contract for initial production inspection. Initial production units shall be numbered and inspected in the sequence listed in table II and as specified in table III.

4.4.3 First article inspection failure. Test item deficiencies during, or as a result of, the first article test, shall be cause for rejection of the items until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of the first article test, shall be prima facie evidence that all items already produced prior to completion of the first article test are similarly deficient unless evidence satisfactory to the contracting officer is furnished by the contractor that they are not similarly deficient. Such deficiencies on all items shall be corrected by the contractor at no cost to the Government.

TABLE II. Sequence of tests.

Test sequence	Paragraph
High temperature	4.6.9.1
Low temperature	4.6.9.1
Humidity	4.6.9.2
Sand	4.6.9.3
Dust	4.6.9.4
Salt fog	4.6.9.5
Chemicals	4.6.9.6
Cleaning spray	4.6.9.7
Fungus	4.6.9.8
Vibration	4.6.9.9
Basic shock	4.6.9.10
Gun firing shock	4.6.9.11
Ballistic shock	4.6.9.12
Burst pressure	4.6.6

TABLE III. Classification of inspections.

Title	Requirement	Inspection	First article	Conformance	
				Examination	Acceptance (100%)
Materials and construction	3.3 thru 3.4 and 3.4.2	4.6.1	X		
Defects	3.4, 3.7 and 3.8	4.6.2	X	X	

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TABLE III. Classification of inspections - Continued.

Title	Requirement	Inspection	First article	Conformance	
				Examination	Acceptance (100%)
Weight	3.4.1	4.6.3	X		
External leakage	3.5.1.1	4.6.4.1	X		X
Internal leakage	3.5.1.2	4.6.4.2	X		X
Proof pressure	3.5.2	4.6.5	X		X
Burst pressure	3.5.3	4.6.6	X		
Heat transfer	3.5.4	4.6.7	X		
Air pressure loss	3.5.5	4.6.8	X		
High temperature	3.5.6.1.1	4.6.9.1	X		
Low temperature	3.5.6.1.2	4.6.9.1	X		
Humidity	3.5.6.2	4.6.9.2	X		
Sand	3.5.6.3	4.6.9.3	X		
Dust	3.5.6.4	4.6.9.4	X		
Salt fog	3.5.6.5	4.6.9.5	X		
Chemicals	3.5.6.6	4.6.9.6	X		
Cleaning spray	3.5.6.7	4.6.9.7	X		
Fungus	3.5.6.8	4.6.9.8	X		
Vibration	3.5.6.9	4.6.9.9	X		
First article		4.6.9.9.1	X		
Basic shock	3.5.6.10	4.6.9.10	X		
First article		4.6.9.10.1	X		
Gun firing shock	3.5.6.11	4.6.9.11	X		
First article		4.6.9.11.1	X		
Ballistic shock	3.5.6.12	4.6.9.12	X		

4.5 Conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the heat exchanger units of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for conformance examination shall be selected in accordance with ANSI/ASQC Z1.4.

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4.5.2 Conformance examinations.

4.5.2.1 Classification of defects. For examination purpose, defects shall be classified as listed in table IV.

TABLE IV. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major:		
101	Assembly, incomplete (see 3.4).	Visual
102	Dimensions affecting interchangeability, out of tolerance (see 3.4 thru 3.4.2).	SIE 1/
103	Identification marking, improper (see 3.7).	Visual
104	Faulty workmanship affecting performance (see 3.8).	Visual
Minor:		
201	Dimensions not affecting interchangeability, out of tolerance (see 3.4 thru 3.4.2).	SIE
202	Faulty workmanship affecting appearance (see 3.8).	Visual

1/ SIE = Standard Inspection Equipment.

4.5.3 Conformance tests.

4.5.3.1 Acceptance tests (100 percent). Each heat exchanger unit shall be subjected to the tests specified in table III.

4.5.3.2 Failure. Failure of any heat exchanger unit to pass any of the specified inspections shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.6 Methods of inspection.

4.6.1 Materials. Conformance to 3.4 through 3.4.2 shall be determined by inspection of contractor records providing proof or certification that materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection

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records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.6.2 Defects. Conformance to 3.4 through 3.4.2, 3.7 and 3.8 shall be determined by examination for the defects listed in table IV. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.6.3 Weight. To determine conformance to 3.4.1, the heat exchanger shall be weighed.

4.6.4 Leakage.

4.6.4.1 External leakage. To determine conformance to 3.5.1.1, the heat exchanger unit shall be connected as shown in figure 4 in the applicable specification sheet. The unit shall be immersed in a water bath and a pneumatic pressure applied to the hot air passages as specified in applicable specification sheet. The external weld seams, side plates, and pans shall be examined to verify that there are no escaping bubbles. Bubbles emerging from the cold air openings are acceptable during this test (see 6.4).

4.6.4.2 Internal leakage. To determine conformance to 3.5.1.2, with the same connection as 4.6.4.1, a pneumatic pressure as specified in the applicable specification sheet shall be applied to the hot air passages. The two shut off valves in the supply line shall be closed and the vent valve opened to ensure complete isolation of the unit from the pressure source. The pressure loss in the first 20 seconds following isolation shall conform to the applicable specification sheet.

4.6.5 Proof pressure. To determine conformance to 3.5.2, the air ports connecting the hot air passages to a pressure source shall be sealed and the cold air ports shall be left open. The heat exchanger unit shall be placed in a safety enclosure as shown in figure 5 in the applicable specification sheet. A pressure as specified in the applicable specification sheet to the hot air passages shall be applied for not less than 5 minutes. At completion the test shall verify that the unit has no visible damage or permanent deformation and conduct the test of 4.6.4.

4.6.6 Burst pressure. To determine conformance to 3.5.3, the air ports shall be sealed while leaving the cooling air ports open. The heat exchanger shall be placed in a safety oven and a thermocouple shall be placed on the heat exchanger skin at a point of high mass as shown in figure 5 in applicable specifications sheet. The oven temperature shall be brought to the specified value in the applicable specification sheet until the unit temperature stabilizes within 5°F (2.8°C) of the oven temperature , but not less than 1 hour. A pressure as specified in the applicable specification sheet shall be applied to the inlet air passages for not less than 5 minutes. Permanent deformation is permissible, however, the unit shall not burst as a result of the applied pressure.

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4.6.7 Heat transfer. To determine conformance to 3.5.4, air flow shall be established in accordance with table VI of the applicable specification sheet, and hot air outlet temperature shall be measured for compliance with table IV.

4.6.8 Air pressure loss. To determine conformance to 3.5.5, the isothermal pressure drop across the inlet air passages shall be measured at flows of 10, 14.7, and 20 pounds per minute (lb/min) (4.5, 6.7 and 9.7 kilograms per minute (kg/min) (±5 percent)) specified in the applicable specification sheet at laboratory ambient temperature. The density correction factor shall be based on average air temperature and average air pressure in the air passages. Pressure loss shall not exceed the values of figure 2 as specified in the applicable specification sheet.

4.6.9 Environmental conditions.

4.6.9.1 Temperature. To determine conformance to 3.5.6.1, the heat exchanger shall be tested as follows:

a. Operation.
   1. Low temperature. The assembly shall be tested in accordance with MIL-STD-810, method 502.3, procedure II at -60°F.
   2. High temperature. The assembly shall be tested in accordance with MIL-STD-810, method 501.3, procedure II at 125°F.

b. Storage.
   1. Low temperature. The assembly shall be tested in accordance with MIL-STD-810, method 502.3, procedure I at -60°F for 24 hours.
   2. High temperature. The assembly shall be tested in accordance with MIL-STD-810, method 501.3, procedure I at 160°F for 48 hours.

4.6.9.2 Humidity. To determine conformance to 3.5.6.2, the heat exchanger shall be placed in a humidity chamber and subjected to the humidity test specified in MIL-STD-810, method 507.3, procedure III. After exposure, the heat exchanger shall pass the test of 4.6.4.

4.6.9.3 Sand. To determine conformance to 3.5.6.3, the heat exchanger shall be tested in accordance with MIL-STD-810, method 510.3, procedure II with exceptions that the sand particles shall be 0.01 to 1 mm diameter and blown by air with a velocity of 3038 feet per minute (ft/min) (30 knots) (55.6 kilometer per hour (km/hr)) against external component surfaces up to 1 hour. Sand concentration shall be 0.3 ±0.2 grams per cubic foot (g/ft^3) (0.0084 ±0.0056 grams per cubic meter (g/m^3)). Following exposure, the heat exchanger shall pass the test of 4.6.4.

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4.6.9.4 Dust. To determine conformance to 3.5.6.4, the heat exchanger shall be tested in accordance with MIL-STD-810, method 510.3, procedure I with exceptions that the dust particles be of 0.0001 to 0.01 mm diameter and $6x10^{-9}$ grams per cubic centimeter (g/cm^3) and blown against external component surfaces at an air velocity of 3544 ft/min (35 knots) (64.8 km/h). Following exposure, the heat exchanger shall pass the test of 4.6.4.

4.6.9.5 Salt fog. To determine conformance to 3.5.6.5, the heat exchanger shall be placed in a test chamber and subjected to the salt fog test specified in MIL-STD-810, method 509.3, procedure I. The salt fog solution shall be 5 percent by weight NaCl in 95 percent by weight of distilled water. The temperature in the exposure zone shall be maintained between 90 and 95°F (32 and 35°C). Fog density shall be approximately three quarts of salt solution per 10 cubic feet (ft^3) (0.3 cubic meter (m^3) chamber volume per 24 hours. After the 48 hour exposure, conduct the test of 4.6.4.

4.6.9.6 Chemicals. To determine conformance to 3.5.6.6, the heat exchanger shall be exposed to the vapors of fuels, hydraulic fluid and cleaning agents in direct contact for a period of 48 hours (see 6.5). During exposure, the unit shall pass the test of 4.6.4.2. After exposure, the unit shall pass the test of 4.6.4.

4.6.9.7 Cleaning spray. To determine conformance to 3.5.6.7, the heat exchanger shall be subjected to a cleaning spray, either water jet or steam spray, using cleaning agents jet rinse, immediately followed by a cold water rinse (see 6.5). The jet spray shall be applied perpendicularly to the surface being cleaned at a distance not less than 1 foot (ft) (0.3 m) from the surface for steam cleaning and not less than 5 ft (1.5 m) from the surface for water jet cleaning, and surface cleaned at a rate of 1 square foot per minute (ft^2/min) (0.09 m^2/min). Nozzle pressure shall be a maximum of 50 pounds per square inch (psi) (758 kPa). Nozzle orifice diameter shall be limited to a maximum of 0.25 in. (6.3 millimeters (mm)) for the water jet. After exposure to the cleaning spray, the heat exchanger shall be visually examined externally for any signs of deterioration and shall pass the test of 4.6.4.

4.6.9.8 Fungus. To determine conformance to 3.5.6.8, the heat exchanger shall be tested in accordance with MIL-STD-810, method 508.4, except that after being exposed to inoculation it shall be followed by exposure to ambient air temperatures of 80 to 84°F (26.7 to 28.9°C) at relative humidities between 95 to 100 percent for a 28 day duration.

4.6.9.9 Vibration. To determine conformance to 3.5.6.9, the heat exchanger shall be mounted using the installed interface configuration for its intended application or equivalent. The heat exchanger shall be tested in accordance with MIL-STD-810, method 514.4, category 8, procedure I, the vibration shall be imposed at a logarithmic sweep rate of 15 minutes per sweep

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while cycling between 5 to 500 to 5 Hz. The test of 4.6.4.2 shall be conducted after each axis as defined in figure 3 of the applicable specification sheet. After vibrating the unit shall pass the test of 4.6.4.

4.6.9.9.1 First article. For first article testing the heat exchanger shall be subjected to a sinusoidal vibration test in accordance with the requirements of table V for a total of 60 minutes in each of three orthogonal axes.

TABLE V. Vibration levels.

Axis	Frequency (Hz)	Amplitude
Vertical	5 to 25 25 to 57 57 to 500	±1 gravity (g) 0.030 inch double amplitude ±5 g
Longitudinal and lateral	5 to 25 25 to 44 44 to 500	±1 g 0.030 inch double amplitude ±3 g

4.6.9.10 Basic shock. To determine conformance to 3.5.6.10, the heat exchanger shall be mounted using the installed interface configuration for its intended application or equivalent. The heat exchanger shall be tested in accordance with MIL-STD-810, method 516.4, procedure I. A total of three half sine wave shock pulses of 30 ±3 g, 11 ±1.1 ms shall be applied in both directions along each of three mutually perpendicular axes for total of 18 shocks (see figure 3 in the applicable specification sheet). At the conclusion of this test, the heat exchanger shall evidence no damage and shall pass the test of 4.6.4.

4.6.9.10.1 First article. For first article testing, the shock amplitude shall be 30 ±3 g, 11 ±1.1 ms.

4.6.9.11 Gun firing shock. To determine conformance to 3.5.6.11, the heat exchanger shall be mounted using the installed interface configuration for its intended application or equivalent, the heat exchanger shall be tested in accordance with MIL-STD-810, method 516.4, procedure VII. A total of three half sine wave shock pulses of 100 ±1 g, 1.5 ±0.2 ms shall be applied in both directions along each of three mutually perpendicular axes for a total of 18 shocks (see figure 3 in the applicable specification sheet). At the conclusion of this test, the heat exchanger shall evidence no damage and shall pass the test of 4.6.4.

4.6.9.11.1 First article. For first article testing peak amplitude shall be 100 ±10 g, 1.5 ±0.2 ms.

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4.6.9.12 Ballistic shock. To determine conformance to 3.5.6.12, the heat exchanger shall be mounted using the installed interface configuration for its intended application or equivalent. The heat exchanger shall be tested in accordance with MIL-STD-810, method 516.4, procedure VII. A total of three half sine wave shock pulses of 200 ±20 g, 0.5 ±0.1 ms shall be applied in both directions along each of three mutually perpendicular axes for a total of 18 shocks (see figure 3 in the applicable specification sheet). At the conclusion of this test, the heat exchanger shall evidence no damage and shall pass the test of 4.6.4.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The three heat transfer components covered by this specification serve to cool and dry M1A1 and M1A2 tank engine bleed air for processing through the NBC system. Although the three units have different names, they are referred to herein by the generic term heat exchanger.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Title, number, and date of the applicable specification sheets.
c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
d. Number and date of the applicable drawing (see 2.2.2).
e. If first article samples are not required (see 3.2).
f. If inspection conditions are other than as specified (see 4.3).
g. If preproduction inspection is required (see 4.4.1).
h. If initial production inspection is not required (see 4.4.2).
i. Packaging requirements (see 5.1).

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6.3 Supersession data. This military specification supersedes the General Dynamics documents listed below:

a. SC-12336771, dated 29 March 1984
b. SC-12336772, dated 29 March 1984
c. SC-12336768, dated 29 March 1984

6.4 Definitions. Air flow terms used in this specification are defined as follows:

a. Hot air. Engine bleed air to be cooled in the heat exchanger.
b. Cold air. Air used as the cooling medium in the heat exchanger.

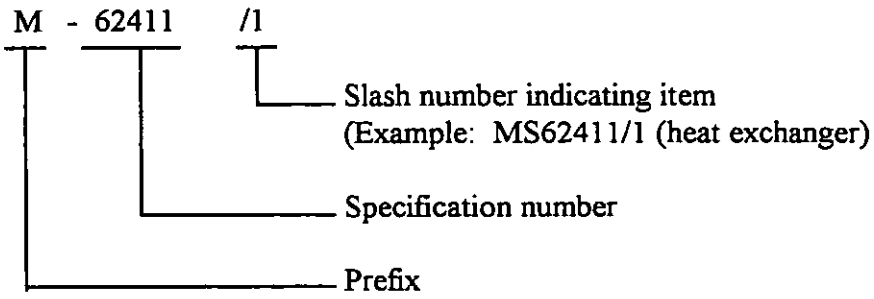
6.5 Chemicals. Chemicals equivalent to the following are to be used in testing as applicable:

Fuel	A-A-52557, MIL-PRF-5624, MIL-G-3056 and MIL-F-16884
Hydraulic fluid	MIL-H-6083 and MIL-H-46170
Clean agent	P-C-437

6.6 Subject term (key word) listing.

Condenser
NBC components
NBC protection
Precooler

6.7 Part Identification or Number (PIN). The PINs to be used for heat exchanger, NBC system acquired to this specification are created as follows:



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6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army - AT

Preparing Activity:
Army - AT

(Project 4130-0395)

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-62411A(AT)	2. DOCUMENT DATE (YYMMDD) 970707

3. DOCUMENT TITLE: HEAT EXCHANGER, NBC SYSTEM GENERAL SPECIFICATION FOR

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE *(Include Area Code)* (1) Commercial (810) 574-8745	(2) AUTOVON 788-8745
c. ADDRESS *(Include Zip Code)* Commander U.S. Army Tank-automotive and Armaments Command ATTN: AMSTA-TR-E/BLUE Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403 Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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